Gravitational field strength

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a gravitational field to predict the force on a test mass.

SUCCESS CRITERIA - I can...

1. Represent field direction with arrows.
2. Use g=F/m and F=mg consistently.
3. Distinguish source field from the mass placed in it.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “field” and “gravity” to explain your model or result. Sentence starters are optional.

field: a description of an influence at each location

gravity: an attractive interaction between masses

radius: distance from the center to the edge of a circle or sphere

01 Warm-up: bring back an earlier idea

An object weighs 29.4 N where g=9.8 N/kg.
Find its mass and its weight in a field of 5 N/kg.

Word help: weight = the gravitational force on an object

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $g = F_g/m = GM/r^2$.
Outside a spherical source of mass M; r is distance from its center. Field direction is toward the source.

Worked reasoning: A 2 kg test object feels 6 N toward a source. The local field is g=6/2=3 N/kg. A 4 kg test object at that point feels 12 N.

Your turn: Draw the field arrow and the two possible force arrows. Which quantity belongs to the location rather than the test mass?

Sentence starter: The ... in my model represents ... . Its label shows ... .

Test mass (kg)	Force (N)	g (N/kg)
2	6	____
4	12	____

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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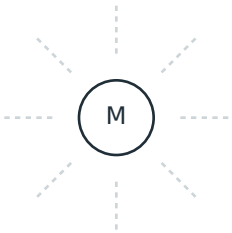
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Draw field arrows at three locations around the source. Which way would a small test mass accelerate?

Sentence starter: The … in my model represents … . Its label shows … .



Add arrowheads to show field direction.

The center marks the source; add arrowheads on the surrounding lines to show field direction.

Marzano check - circle: 1 2 3

04 Apply the physics

A 0.20 kg test mass experiences a 0.60 N gravitational force.
Find the field strength. The source has GM = 12 m^3/s^2; find the test mass's center-to-center distance from the source.

Sentence starter: I found … [unit]. This value means … in this situation.



Source: GM = 12 Test mass: 0.20 kg

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student says a heavier test mass falls faster because it feels more gravitational force.
Use F=mg and a=F/m to respond.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A 4 kg block slides down a 30° ramp with 11.6 N friction up the ramp.
Use g=9.8 m/s².
Find along-ramp acceleration and normal force.
Draw the forces on the chosen object and label their sources.

Sentence starter: I found ... [unit]. This value means ... in this situation.

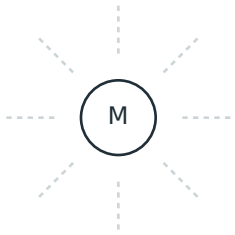
Add and label force arrows.

Marzano check - circle: 1 2 3

07 Exit check

Outside a spherical source, field is 8 N/kg at r.
Find field at 2 r and force on 0.50 kg there.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add arrowheads to show field direction.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

An exterior field is 8 N/kg at distance r from a spherical source. Where is it 2 N/kg? What force acts there on a 0.25 kg mass?

Sentence starter: I notice ... . This supports ... because ... .

Location	Test mass (kg)	Field (N/kg)	Force (N)
1	____	____	____
2	____	____	____

Marzano check - circle: 1 2 3